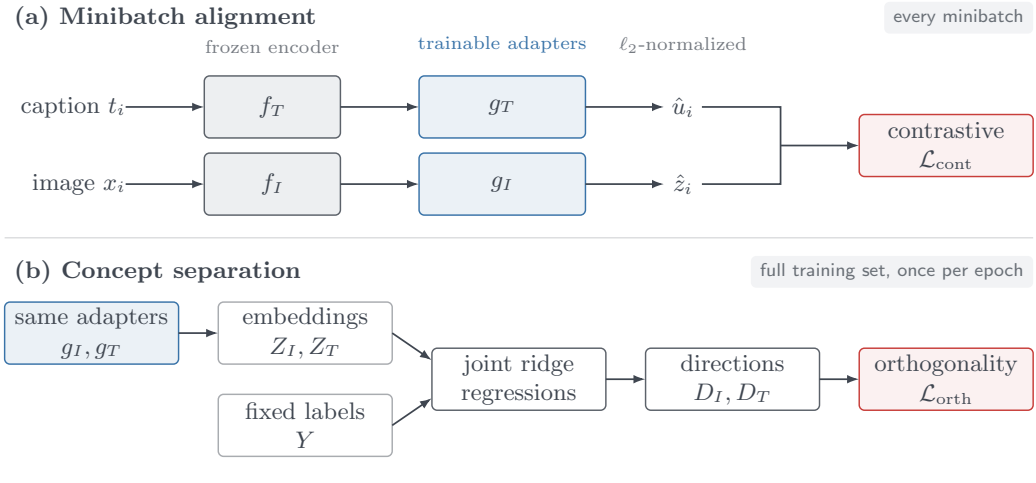

(a) Minibatch alignment
every minibatch
frozen encoder
trainable adapters
$\ell_2$-normalized
caption $t_i$
image $x_i$
$f_T$
$f_I$
$g_T$
$g_I$
$\hat{u}_i$
$\hat{z}_i$
contrastive
$\mathcal{L}_{\text{cont}}$
(b) Concept separation
full training set, once per epoch
same adapters
$g_I, g_T$
embeddings
$Z_I, Z_T$
fixed labels
$Y$
joint ridge
regressions
directions
$D_I, D_T$
orthogonality
$\mathcal{L}_{\text{orth}}$